SPECTRAL DECOMPOSITION

The Spectral Theorem

Every real symmetric matrix is orthogonally diagonalizable.

Real symmetric A

$A = A^T$

GUARANTEES →

Three facts

- All eigenvalues are real; no complex eigenvalues can appear
- Eigenvectors for distinct eigenvalues are orthogonal: $\lambda_i \neq \lambda_j \implies \mathbf{q}_i \cdot \mathbf{q}_j = 0$
- There are n orthonormal eigenvectors even when eigenvalues repeat: Gram–Schmidt inside each repeated eigenspace

TOGETHER GIVE →

Orthogonal diagonalization

$A = QDQ^T$

- Q: an orthonormal basis of eigenvectors for $\mathbb{R}^n$
- Exists for every real symmetric A